

CS 502Midterm preparationQuestion:1

Criterion of edge selection in greedy algorithm for finding a MST in any graph?

Answer:

Select the minimum weight edge that does not form a cycle or connects visited to unvisited node.

Question:2

What is the basic functionality of counting sort algorithm?

Note: List down at least two.

Answer:

- 1) It counts how many times each value appears in the input array.
- 2) It calculates the rank of each value to place it directly in the sorted array.

Question: 3

write a pseudo code of Radix sort algorithm for d digits long numbers.?

Answer:

Here is the algorithm that sorts $A[1...n]$ where each number is d digits long

Radix-Sort (array A , int n , int d)

- 1) for $i \leftarrow 1$ to d
- 2) do stably sort A w.r.t i^{th} lowest order digits.

Question: 4

How divide and conquer strategy applied while designing an algorithm.?

Answer:

Divide: The problem into a small number of pieces.

Conquer: Solve each piece by applying divide and conquer to it recursively.

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Combine: the piece together into a global solution.

Question: 5

What is algorithm why we use it and its properties?

Answer:

An algorithm is a step by step procedure to solve a problem. We use it to get a fast, correct, and efficient result in a limited time.

Properties an algorithm:

- 1) Input 2) Output 3) Definiteness
- 4) Finiteness 5) Effectiveness

Question: 6

How many number of entries in matrix in edit distance?

Answer:

The edit distance matrix has $(m+1) \times (n+1)$ entries, where m and n the length of the two strings. Each

cell stores the minimum operation to convert one substring to another, used in dynamic programming.

Question: 7

Asymptotic notations equivalent or not? Give reason.

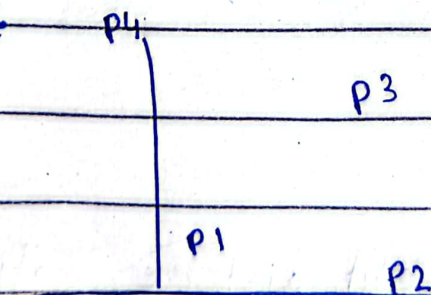
Answer:

No, asymptotic notations are not equivalent. Each notation (Big-O, Omega, Theta) describes different aspects of algorithm performance best, worst, or tight bounds.

Question: 8

Draw a graph of maximal points?

Answer:



Question: 9

What is bin sort algorithm?

Answer:

- i) Divides elements into several bins based on their value range.
- ii) Each bin is sorted individually.
- iii) Finally all bins are combined to get the sorted list.

Question: 10

What is Asymptotic Notation?

Answer:

Asymptotic notation describe the time or space complexity of an algorithm as input size grows. It shows the algorithm's performance in the long run.

- i) Big ii) Omega iii) Theta

Question: 11

What is the time complexity of brute force and plane sweep algorithm?

Answer:

Date: _____

Brute Force:

$O(n^2)$ checks all possible pairs slow.

plane Sweep:

$O(n \log n)$ uses sorting + efficient scanning faster.

Question: 12

Why average time case complexity of Quick Sort not distribution of input?

Answer:

Because Quick Sort average case depends on the random position of pivots not the input order. It assumes all pivot positions are equally likely.

Question: 13

Apply merge sort on following numbers?

7, 9, 1, 5, 3, 8, 4, 2, 6

Answer:

Date: _____

$[1, 9, 15, 3, 8, 4, 2, 6]$

$[1, 9, 15] [3, 8, 4, 2, 6]$

$[1, 9] [15] [3, 8] [4, 2, 6]$

$[1] [9] [15] [3] [8] [4] [2] [6]$

Final ans:

1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Question: 14

Define between divide & conquer and dynamic programming?

Answer:

Divide and Conquer:

Solves a problem by dividing it into smaller sub problem, solving them independently, and combine the result.

Dynamic Programming:

Solve problem by breaking into sub problems storing results to avoid repeated work.

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Question: 15

Define heap?

Answer:

A heap is a tree based data structure where each parent node is either greater than or less than its child nodes. It is used in heap sort and priority queues.

Question: 16 Find

Difference between graph and free. Find the value G^t ?

Answer:

Graph:

Nodes connected by edges.

Free:

No extra restrictions (like free graph)

G^t - Transpose of matrix G (rows and columns swapped).

Question: 17 Final

What is Dijkstra algorithm?

Answer:

Dijkstra algorithm is a method to find the shortest path from a source node to all other nodes in a graph.

Question: 18

How do you create edge in a course graph?

Answer:

To create edge in a course graph, draw a directed edge from each prerequisite course to the course that depends on it.

Question: 19 Final

What are DFS and BFS in graph traversal?

Answer:

DFS (Depth-First Search) explores

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as far as possible along each branch before backtracking.

BFS (Breadth First Search) explores all neighbors at the current depth before moving to the next level.

Question: 20 Find

What are the 4 steps of Dynamic programming?

Answer:

- 1) Define the problem in terms of subproblem.
- 2) Write a recurrence relation.
- 3) Compute the solution to subproblem.
- 4) Build the final solution from the computed values.

Question: 21

Explain the working of insertion Sort briefly.?

Answer:

Insertion Sort builds a sorted list by picking one element at

a time and inserting it into the correct position in the already sorted part.

Question: 22

What is the function of greater or equal in programming?

Answer:

It compare two values and returns true if the first value is greater than or equal to the second otherwise, it return false.

Question: 23

Two points maxima with graph.

Answer:

Two highest points (local maxima) in a graph.

Question: 24

The following Statement summation from represent quick sort so what will be outcome after solving the recurrence in terms of asymptotic notation

$$\leq = \frac{(n+1)(n+2)}{2}$$

Answer:

$$(n+1)(n+2)$$

$$n^2 + 3n + 2$$

$$\Theta(n^2) \text{ As,}$$

Question: 25

If $i \neq j$ and $P[i].x$ are two points in 2 dimension space and the following condition are true then which point is maximal. If $(i \neq j)$ and $(P[i].x \leq P[j].x)$ and $(P[i].y \leq P[j].y)$

Answer:

$P[i]$ is maximal because it dominates $P[j]$ in both x and y coordinate.

Question: 26

Find the statement is true.

- 1) The worst case time is complexity of the selection sort is $O(N^2)$
- 2) The worst case time is complexity of the merge is $O(\log n)$

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Answer:

The worst case time complexity of the selection sort is $O(N^2)$ is true.

Question: 27

Write down the difference between Divide Conquer and Dynamic Programming technique?

Answer:

Divide and Conquer:

Divide and Conquer split the problem into independent subproblem and combine their result.

Dynamic Programming:

Dynamic Programming handles overlapping subproblem and store result to avoid repetition.

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Question: 28

Write the summation form of a quadratic series using theta notation?

Answer:

A quadratic series can be written as:

$$\sum_{i=1}^n i^2 = \Theta(n^3)$$

Question: 29

How does the bubble sort algorithm work?

Answer:

Bubble Sort works by repeatedly comparing adjacent elements and swapping them if they are in the wrong order. This process continues until the list becomes completely sorted.

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Question: 30

What is plagiarism detection using edit distance?

Answer:

Plagiarism detection with edit distance compares two texts by measuring how many edits are needed to change one text into the other. A low edit distance indicates high similarity, which helps detect copied content.

Question: 31

What are the two main criteria to measure or analyze an algorithm?

Answer:

- i) Time complexity
- ii) Space complexity

Question: 32

What is the difference between a computer program and an algorithm?

Answer:

Algorithm:

An algorithm is a step by step solution or procedure to solve a problem.

Computer program:

A computer program is the actual code written in a programming language to implement an algorithm.

Question: 33

Which of the following are simple comparison based sorting algorithms?

Answer

- i) Merge sort ii) Bubble sort
- iii) Insertion sort iv) Selection sort

Question: 34

Sort the array [7, 2, 5, 1, 9]
using quick sort.?

Answer:

Date: _____

Pivot = 7

Left [2, 5, 1]

Right [9]

Sort left [1, 2, 5]

Combine [1, 2, 5, 7, 9] ✓

Question: 35

How does bubble sort works?

Answer:

Bubble sort compare adjacent element and swaps them if they are in the wrong order. This process repeats until the list is sorted.

Question: 36

What are the main function of Counting sort?

Answer:

Counting Sort first counts how many times each element appears in the

Date: _____

array. Then it calculates the cumulative count to find each element correct position. Finally it place the element in a new array to produce a sorted list.

Question: 37

Running time of $3n^2 + 7n - 12$
find the constant C and n not?

Answer:

$$T(n) = 3n^2 + 7n - 12$$

$$C = 4 \quad \text{then solve } 3n^2 + 7n - 12 \leq Cn^2$$

$$n^2 - 7n + 12 \geq 0$$

$$(n-3)(n-4) \geq 0 \quad n \geq 4$$

$$O(n^2), C = 4, n_0 = 4$$

Question: 38

Write the pseudocode of Heap Sort algorithm?

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Answer:

Heap Sort (A)

Build Max Heap (A)

for $i = \text{length}(A)$ down to 2

Swap $A[1]$ and $A[i]$

$n = n - 1$

Max Heap if $y(A, 1)$

Question: 39

Write any two basic functionalities of Count Sort.?

Answer:

i) Counting element ii) Cumulative account

Counting element:

Counts the number of occurrences of each element.

Cumulative account:

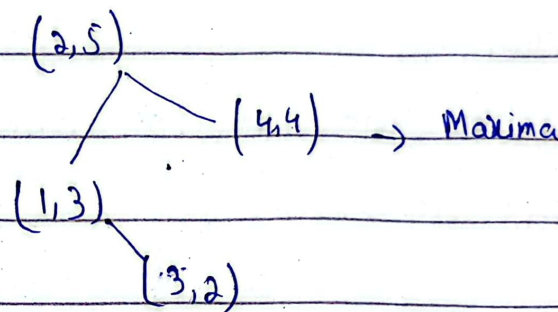
Calculate cumulative counts to determine the position of each element in the sorted array.

Question: 40

Explain 2D maxima problem with the help of a pictorial diagram?

Answer:

In a set of 2D points a point (x, y) is called a maxima if no other point has both coordinates greater than or equal to it.



Question: 41

Sort the array $(50, 20, 60, 10, 40)$ using quick sort taking the first element as pivot?

Answer:

Pivot = 50

Partition = $(20, 10, 40, 50, 60)$

Sort left side $(20, 10, 40)$ $(10, 20, 40)$

(10, 20, 40, 50, 60) A,

Question: 42

Explain the conquer strategy of Quick Sort in three steps?

Answer:

i) Partition ii) Recur iii) Combine

Partition:

Choose a pivot and divide the array into two subarrays.

Recur:

Apply Quick Sort recursively on the left and right sub arrays.

Combine:

Combine the sorted subarrays they form a fully sorted array.